FINAL EXAM
MATH 54

1. Let $A = \begin{bmatrix} 1 & 2 \\ 8 & 1 \end{bmatrix}$.

(a) Compute eigenvalues of A.

(b) Find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

2. Let A be an invertible 3×3-matrix. Prove the following statements:

(a) If all entries of A and A^{-1} are integers then $\det A = \pm 1$.

(b) If all entries of A are integers and $\det A = \pm 1$ then all entries of A^{-1} are integers.

3. **For this problem the explanations are not required. Full credit will be given for correct answers.**

Let V be a 4-dimensional subspace in $\mathbb{R}^{20}$, $T : \mathbb{R}^{20} \to \mathbb{R}^{20}$ defined by the formula

$$T(\mathbf{y}) = \operatorname{proj}_V \mathbf{y},$$

where proj_V denotes the orthogonal projection onto V. Assume that A is the standard matrix of T.

(a) Find all eigenvalues of A.

(b) Compute the characteristic polynomial for A.

(c) Find the rank of A.

(d) Find the dimension of $\operatorname{Nul} A$.

(e) Is A symmetric?

4. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$.

(a) Find the singular values of A.

(b) Find the singular value decomposition $A = U\Sigma V^T$.

5. Let A be a matrix.

(a) Prove that $\operatorname{Nul} A^T A = \operatorname{Nul} A$. (Hint: use the dot product.)

(b) Prove that the rank of A equals the rank of $A^T A$.

6. Solve the initial value problem

$$y'' + 4y = e^t, \quad y(0) = 1,\ y'(0) = 2.$$

7. Consider the homogeneous system of equations

$$\mathbf{x}'(t) = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \mathbf{x}(t).$$

(a) Find a fundamental matrix.

(b) Solve the system for the initial value $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

8. Determine whether the following vector functions

$$\begin{bmatrix} \sin x \\ \sin x \end{bmatrix}, \begin{bmatrix} \sin x \\ \cos x \end{bmatrix}, \begin{bmatrix} \cos x \\ \cos x \end{bmatrix}$$

are linearly dependent or linearly independent on the interval $(-\infty, \infty)$.

9. Consider the function $f(x)$ on the interval $[-\pi, \pi]$ defined by

$$f(x) = \begin{cases} 0 \text{ if } -\pi \leq x \leq 0 \\ 1 \text{ if } 0 < x \leq \pi \end{cases} .$$

(a) Write the Fourier series for $f(x)$.

(b) Denote by $g(x)$ the periodic function to which this Fourier series converges. Find the values $g(0)$, $g(\frac{\pi}{2})$ and $g(\pi)$.

(c) Write $g(\frac{\pi}{2})$ as a sum of infinite series using (a) and (b).